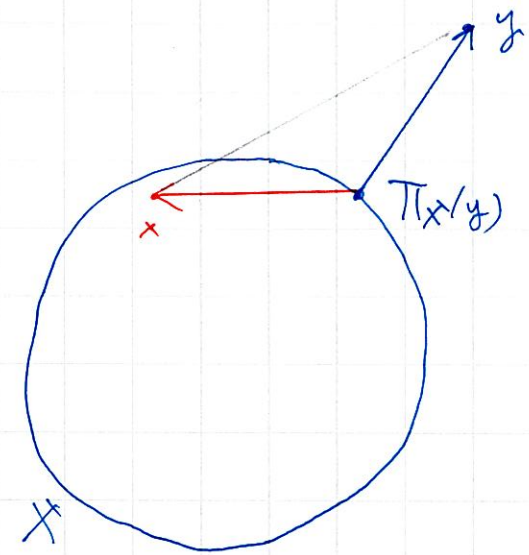
Two Properties of Projection.

- $(x - \Pi_X(y))^t \cdot (y - \Pi_X(y)) \leq 0.$

  for X convex, $y \in \mathbb{R}^n$, $x \in X$.



- For X convex, $y \in \mathbb{R}^n$, $x \in X$, we have

$$\|x - y\|^2 \geq \|y - \Pi_X(y)\|^2 + \|x - \Pi_X(y)\|^2$$

Projected gradient Descent:

$$y^{k+1} = x^k - \gamma \nabla f(x^k) \quad , \quad x^{k+1} = \Pi_X(y^{k+1})$$

- We will use the following preliminary analysis to give a convergence result for projected gradient descent

Let $x^* = \arg\min_{x \in X} f(x)$

- The error of each iteration is $f(x^k) - f(x^*)$. We want to bound error
- By first-order characterization of convexity

$$f(x^*) \geq f(x^k) + \nabla^t f(x^k) \cdot (x^* - x^k) \Rightarrow$$

$$f(x^k) - f(x^*) \leq \nabla^t f(x^k) \cdot (x^k - x^*)$$

- So it is enough to upper bound $\nabla^t f(x^k)(x^k - x^*)$
- Let $g^k = \nabla f(x^k)$

- $\nabla^t f(x^k) \cdot (x^k - x^*) = (g^k)^t \cdot (x^k - x^*) = \frac{1}{\gamma} \cdot (x^k - y^{k+1})^t (x^k - x^*)$

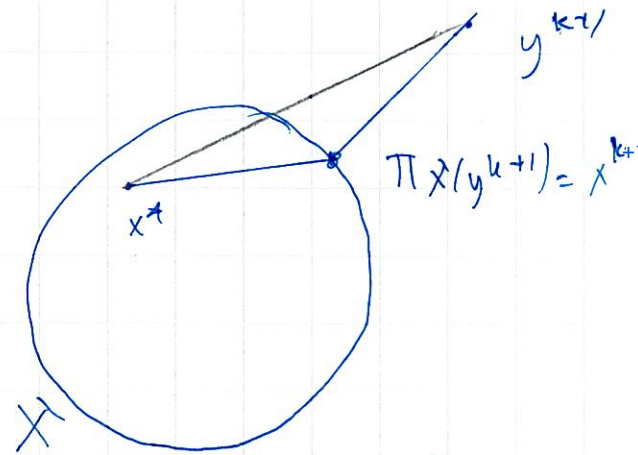
$$\overset{\star}{=} \frac{1}{2\gamma} \left(\|x^k - y^{k+1}\|^2 + \|x^k - x^*\|^2 - \|(x^k - y^{k+1}) - (x^k - x^*)\|^2 \right)$$

$$= \frac{1}{2\gamma} \left(\gamma^2 \|g^k\|^2 + \|x^k - x^*\|^2 - \|y^{k+1} - x^*\|^2 \right) \quad (\dagger)$$

$\star$ follows by $\|a-b\|^2 = \|a\|^2 + \|b\|^2 - 2a^t b$ so,
$a^t b = \frac{1}{2} \left(\|a\|^2 + \|b\|^2 - \|a-b\|^2 \right)$

- We know from second property of projection

$$\|y^{k+1} - x^*\|^2 \geq \|x^{k+1} - x^*\|^2 + \|y^{k+1} - x^{k+1}\|^2$$
$$\geq \|x^{k+1} - x^*\|^2 \quad (\ddagger)$$



- By $(\dagger)$ and $(\ddagger)$, we get

$$\nabla f^t(x^k)(x^k - x^*) \leq \frac{\gamma}{2} \|g^k\|^2 + \frac{1}{2\gamma} \left(\|x^k - x^*\|^2 - \|x^{k+1} - x^*\|^2 \right)$$

- Add this inequality from $k=0$ to $k=K-1$ to get:

$$\sum_{k=0}^{K-1} \nabla f^{+}(x^k)\,(x^k - x^*) \le \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma}\left(\|x^0 - x^*\|^2 - \|x^K - x^*\|^2\right)$$

$$\le \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma}\|x^0 - x^*\|^2$$

- Thus, we get

$$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \le \sum_{k=0}^{K-1} \nabla f^{+}(x^k)\,(x^k - x^*) \le \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma}\|x^0 - x^*\|$$

Convergence of Projected Gradient Descent with Bounded gradients

Thm. Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex and X be a convex set

$$x^* = \arg\min_{x \in X} f(x).$$

Assume $\|x^0 - x^*\| \le R$ and $\|\nabla f(x)\| \le B$ for all $x \in X$

Then, choosing the step size as $\gamma = \frac{R}{B\sqrt{K}}$, then iterations of projected gradient descent satisfy

$$\frac{1}{K} \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \le \frac{RB}{\sqrt{K}}.$$

Pf: By our preliminary analysis: $\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \le \frac{\gamma}{2} K \cdot B^2 + \frac{1}{2\gamma} R^2 \Rightarrow$

$$\frac{1}{K} \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \le \frac{\gamma}{2} B^2 + \frac{1}{2\gamma} \frac{R^2}{K} = \frac{1}{2} \cdot \left(\frac{R}{\cancel{B\sqrt{K}}} \cdot B^{\cancel{2}} + \frac{B\cancel{\sqrt{K}}}{\cancel{K}} \frac{R^{\cancel{2}}}{\cancel{\sqrt{K}}} \right) = \frac{RB}{\sqrt{K}}$$

done!

- Note that $\|\nabla f(x)\| \leq B$ for all $x \in X$ is not a horrible assumption when X is bounded.

Computing projections

- At each iteration of projected gradient descent, we need to compute $\Pi_X(y^{k+1})$.

- Computing this projection is equivalent to solving:

$$\min_{x \in X} \sum_{i=1}^{n} (x_i - y_i^{k+1})^2$$

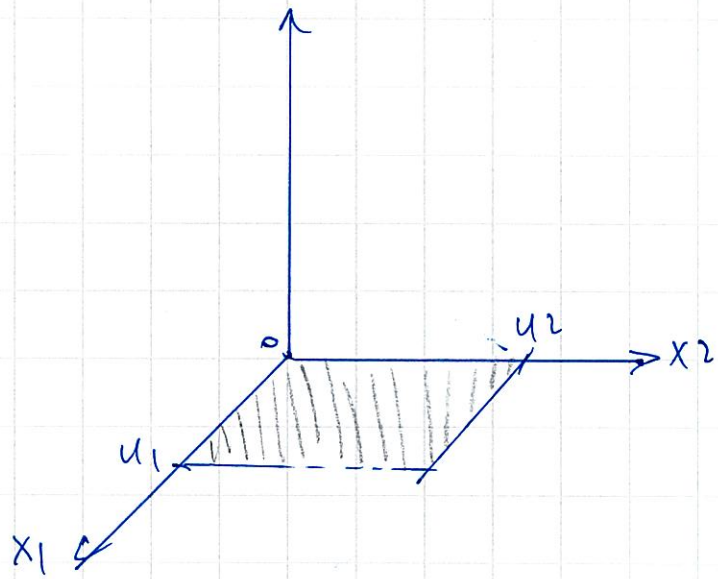
- This is yet another constrained optimization problem! By exploiting the structure of the norm function $\sum_{i=1}^{n} (x_i - y_i^{k+1})^2$, we can compute projections onto X for many different feasible sets X.

- Here are some feasible sets X for which projections are simple to compute.

Projections on a Box

Consider the case $X = \{ x \in \mathbb{R}^n : 0 \le x_i \le u_i \quad \forall i = 1, \dots, n \}$

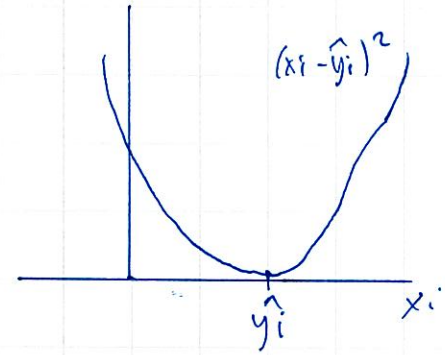
For example, if $n = 2$



- We want to find $\Pi_X(\hat{y})$, where $\hat{y}$ is a fixed point.
- We want to solve $\min_{x \in X} \|x - \hat{y}\|^2$

- We want to solve $\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$ st

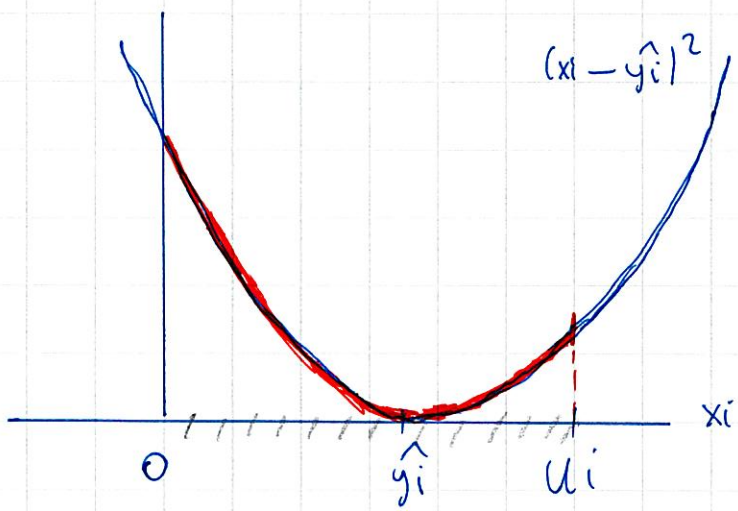
$$0 \le x_i \le u_i \quad \forall i = 1, \dots, n$$



- This problem is equivalent to $\sum_{i=1}^{n} \min (x_i - \hat{y}_i)^2$ st $0 \le x_i \le u_i$

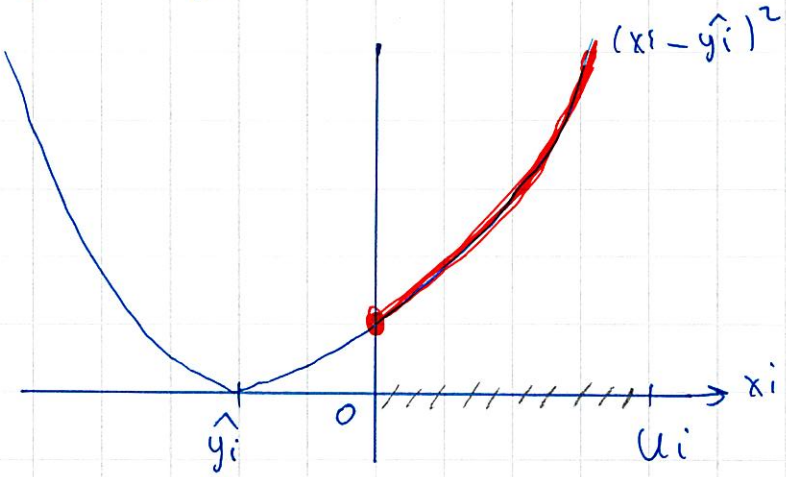
- Let's look at 3 cases:

- Case 1 $\quad 0 \le \hat{y}_i \le u_i$



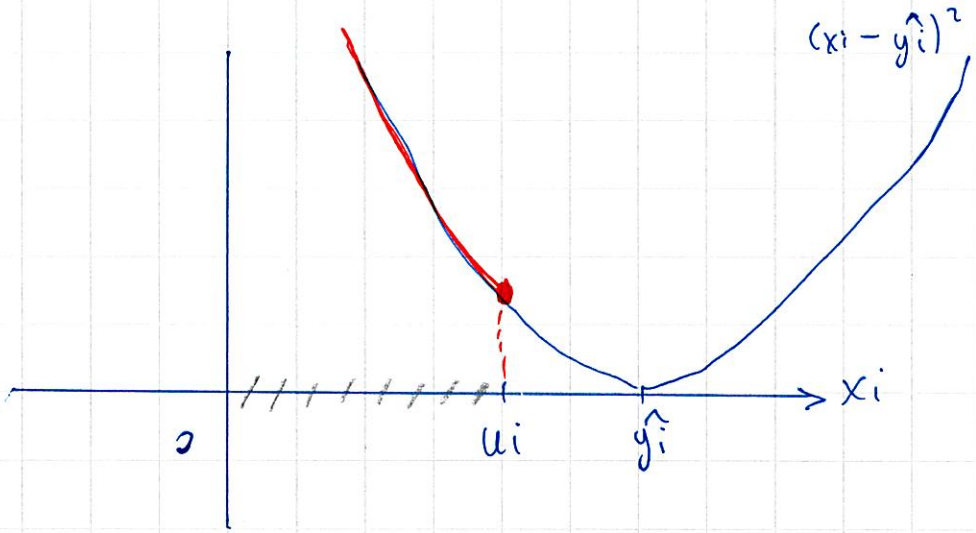
$$\hat{y}_i = \arg\min (x_i - y_i)^2 \quad \text{st} \quad 0 \le x_i \le u_i$$

- Case 2 $\hat{y}_i \leq 0$



$$0 = \underset{\text{st}}{\operatorname{argmin}} \; (x_i - \hat{y}_i)^2 \quad 0 \leq x_i \leq u_i$$

Case 3: $\hat{y}_i \geq u_i$



$$u_i = \underset{\text{st}}{\operatorname{argmin}} \; (x_i - \hat{y}_i)^2 \quad 0 \leq x_i \leq u_i$$

Thus, $\underset{\text{st } 0 \le x_i \le u_i}{\arg\min} (x_i - \hat{y}_i) = \min\{u_i, [\hat{y}_i]^+\}$,

where $[a]^+ = \max\{0, a\}$

- Thus, projection of $\hat{y}$ onto X is given by $\pi = (\pi_1, \dots, \pi_n)$, where

$$\pi_i = \min\{u_i, [\hat{y}_i]^+\}$$